

# Passivity-Based Design for Event-Triggered Networked Control Systems

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**Abstract**—In this paper, we introduce a passivity-based design framework for event-triggered networked control systems. We consider the effects of network-induced time delays, signal quantization, and data loss in communication links between the plant and controller and show  $L_2$ -stability and robustness for the control design. We introduce simple asynchronous triggering conditions that do not rely on the exact knowledge of the systems' dynamics and are located on both sides of the communication network. This leads to a great decrease in the communication rate between systems. Additionally, we show lower bounds on interevent time intervals for the triggering conditions and characterize the design's robustness against external disturbances. We illustrate the relationship among stability, robustness, and passivity levels of the plant and controller. Moreover, we analyze our design's robustness against packet dropouts. Finally, we calculate the passivity levels for the entire event-triggered networked control system. This is beneficial in design of compositional networked control systems. Our results are design-oriented. By following our proposed framework, the designer can characterize clear tradeoffs among passivity levels, design parameters, time delays, effects of signal quantization and triggering conditions, stability, robustness, and performance, and make design decisions accordingly.

**Index Terms**—Networked control systems (NCSs), packet dropouts, passivity, quantization, time delays.

## I. INTRODUCTION

NETWORKED control systems (NCSs) provide the possibility for the remote control of systems that are located in different regions. Under this distributed control framework, a shared multipurpose communication network offers the platform on which controllers and plants exchange information to further the overall goal of a large-scale control system. NCSs offer many advantages such as flexibility, low cost of design, efficiency, and ease of maintenance. As a result, NCSs have

recently enjoyed the research spotlight in fields such as control and communication theory [1]. NCS designs and their benefits are utilized in a large range of areas such as unmanned aerial vehicles [2], remote surgery [3], and mobile sensor networks [4]. A comprehensive survey on recent developments in this field can be found in [5].

Utilizing a communication network between the controller and plant—in contrast to traditional dedicated connections—to achieve control goals brings about several challenges such as possible temporary communication failures due to packet dropouts, or limited communication capabilities caused by implementation of a shared band-limited network, time delays, and signal quantization. Delays in NCSs' communication links can be highly variable due to variant network access times and transmission delays. Some results in the literature dealing with the problem of stability and time delays propose upper bounds for the maximum allowable delays (MAD) and maximum allowable transmission interval (MATI) [6], [7] with the requirement that MAD should be upper bounded by MATI. In [8], Gao *et al.* consider the delays in NCSs and produce a Lyapunov-based method with a desired  $H^\infty$  performance index. Additional works dealing with time delays in NCSs are given in [9]–[11]. The problems of packet dropouts and signal quantization have been addressed in the literature as well. In [12] and [13], Tsumura *et al.* discuss the tradeoffs between quantization, packet dropouts, and stability. Fu [14] provides an analysis of the negative effects of quantization on stability and estimation in NCSs. An linear matrix inequality (LMI)-based approach for dealing with packet dropouts and delays is proposed in [15]. Similar works addressing the problem of information loss in NCSs are given in [16] and [17]. Additionally, in order to overcome the challenges mentioned above, the design of NCSs under multichannel communication links with varying communication capabilities has been investigated in the literature, and its benefits in comparison to other methods have been discussed [18], [19].

Classical NCS designs rely on the traditional periodic task of control. This leads to quite conservative approaches that require an abundance of resources in terms of computation capabilities and network usage [20], [21]. Event-triggered control with the main purpose of creating a better balance among control, communication, and computational load has been proposed as an efficient alternative. Under an event-based structure, information between the plant and controller is exchanged only when necessary—namely, when a certain controlled value deviates from its desired value beyond a certain threshold. Given this framework, the control actions for maintaining stability are only applied at triggering instances [22]–[28]. Most stability-focused works in the field of event-triggered NCSs can be divided into two categories:

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- 1) Input-to-state stability, where the connection between stability and states of the system is explored. This approach usually relies on the full knowledge of the states, which may be unattainable in some cases [25], [29]–[31].
- 2) Stability in terms of input–output finite-gain stability [32]–[35].

In the present work, a new approach for designing finite-gain stable event-triggered NCSs is introduced and analyzed.

### A. Main Contributions

In this paper, we show input–output finite-gain stability for our proposed event-triggered networked control design. A large share of works in the networked control literature assumes perfect communication links with no time delays and fails to consider the effects of signal quantization and packet dropouts. In our work, we attempt to address a more realistic problem by modeling constant or time-varying network-induced time delays and signal quantization and considering their negative effects on the stability of NCSs. Under our proposed design, finite-gain  $L_2$ -stability is guaranteed and a simple transformation matrix  $M$  is applied on the plant's side of the communication links to deal with the presence of signal quantization and time delays with bounded rates of change. The delays existing in both communication links, from the plant to controller and from the controller to plant, are considered. As is apparent from our results, time delays, signal quantization, and packet dropouts do negatively affect the stability and performance of NCSs. A large body of output-based networked control works in the current literature such as the ones presented in [7], [31], and [35] do not consider any of these effects. As a result, these works will require further analysis when they are applied to more realistic practical settings. For example, the results in [7] only hold based on the assumption that the transmission network delays are shorter than the time intervals between each set of triggering instances. This may be quite unrealistic in practice. In the present paper, we will show that our NCS design can tolerate network-induced time delays that are larger than the time interval between triggering instances and that these delays will not affect our stability results as long as the proposed conditions are met. In this respect, our results seek to be more comprehensive compared to the existing results in the literature in the sense that we propose a design-based control framework for NCSs that takes into account the negative effects imposed on NCSs by time delays, signal quantization, and packet dropouts.

In our work, we take advantage of the theory of dissipativity and passivity to design stable and robust event-triggered NCSs. Passivity and dissipativity characterize the energy consumption of a dynamical system. Passivity is preserved under parallel and feedback interconnections [36]. Passivity implies stability under mild assumptions [36], [37] making it a great candidate for designing compositional large-scale NCSs. In this respect, our work is similar to [32]; however, Yu and Antsaklis [32] do not address the problem of packet dropouts and do not consider the robustness issues in regard to external disturbance and noise. Additionally, our proposed event-triggered networked design is more efficient, does not require the “dead-band” unit, and consists of asynchronous event-triggered implementations that are placed on both sides of the communication network. This leads to a large decrease in the communication load imposed on the band-limited network. Specifically, we will show that we can greatly decrease the rate of communication from the controller

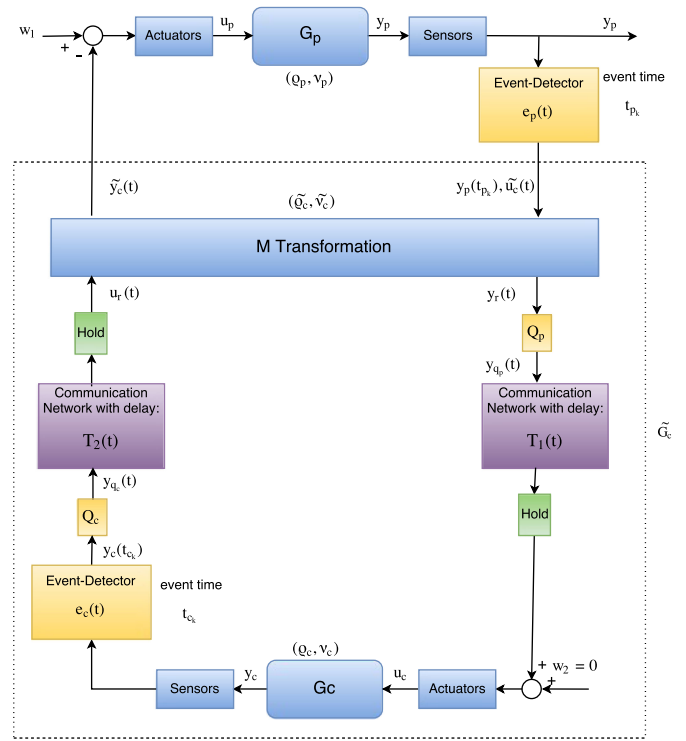


Fig. 1. Networked control system interconnection of two IF-OPF systems with quantization, time-varying delays, and event-detectors on both sides.

to plant. Compared to [32], our results cover a larger class of passive systems as we do not impose any restrictions on the relationship between passivity indices of the controller and plant. This also leads to less conservative stability and triggering conditions. Furthermore, we calculate the passivity indices for the entire event-triggered NCS presented in Fig. 1. This is beneficial because as mentioned a feedback or feed-forward interconnection of passive subsystems is also passive and preserves stability [36]. In application, in cases where compositionality is required such as the design and control of cyber-physical systems, one can follow our proposed framework and design larger compositional NCSs from smaller stable event-triggered NCSs with predesigned specific passivity indices.

We will address the robustness issues in regard to packet dropouts and disturbance. Since our triggering conditions rely on the outputs of the controller and plant, it is necessary to show a positive lower bound for the event instances to ensure that the main purpose of the event-triggered framework—reducing the communication rate—is met. This is to ensure against a continuous exchange of information between subunits or what is known as “Zeno” behavior. Mathematically, we show that the occurrence of an infinite number of event instances in any finite time interval or the “Zeno phenomenon” is not possible. We show lower bounds for interevent time intervals based only on innate conic properties of input feed-forward output feedback passive (IF-OPF) systems. Unlike most approaches in the literature (see [25] and [32]), our method requires no additional assumptions on the dynamics of the system. For instance, the Zeno analysis in [32] may not hold for a large class of dynamical systems as the authors impose further restrictions—in addition to stability restrictions—on the dynamics of the system and the relationship between passivity indices of subsystems.

As mentioned, we propose a design-based approach applicable in practical settings. We consider a large class of systems known as IF-OFP systems and derive triggering conditions based on a passivity-based approach that covers a large class of systems with negative or positive known passivity indices. Our design includes simple event-triggering conditions that are located on both sides, the plant's output and controller's output. There are two benefits to this approach: first, the designer will be able to characterize clear tradeoffs among the communication rate, stability, robustness, and performance, and make design decisions accordingly; second, the presence of asynchronous triggering conditions on both sides of the network will significantly decrease the communication load on the shared band-limited network. Our work may be applied to linear and nonlinear systems and does not rely on the exact knowledge of subsystems' dynamics; only an access to inputs and outputs of each system is sufficient. Our results clarify the robustness of each triggering condition against external disturbance. We analyze the performance of our design under packet dropouts. We show a tradeoff among the design parameters and number of allowed consecutive packet dropouts.

This paper is organized as follows. Section II presents the preliminarily mathematical definitions. The detailed structure of the NCS framework and problem statement are clearly stated in Section III. Section IV presents the results of the design of the event-triggering conditions, the proposed transformation matrix, and  $L_2$ -stability. This section also presents the results in regard to the passivity levels of the entire NCS and their relationship with design parameters. In Section V, analyses of interevent time intervals based on conic properties of IF-OFP systems and robustness issues with respect to packet dropouts and external disturbance are presented. An illustrative example is presented in Section VI. Section VII presents a summary of what was achieved and concludes the paper.

## II. MATHEMATICAL PRELIMINARIES

Consider the linear or nonlinear dynamical system  $G$

$$G : \begin{cases} \dot{x}(t) = f(x(t), u(t)) \\ y(t) = h(x(t), u(t)) \end{cases} \quad (1)$$

where  $f$  and  $h$  are Lipschitz functions,  $x(t) \in X \subset \mathbb{R}^n$ ,  $u(t) \in U \subset \mathbb{R}^m$ , and  $y(t) \in Y \subset \mathbb{R}^m$  are, respectively, the state, input, and output of the system, and  $X$ ,  $U$ , and  $Y$  are the state, input, and output spaces.

**Definition 1:** (see [38]) The supply rate  $\omega(u(t), y(t))$  is a well-defined supply rate, if for all  $t_0, t_1$ , where  $t_1 \geq t_0$ , and all solutions  $x(t) \in X$ ,  $u(t) \in U$ , and  $y(t) \in Y$  of the dynamical system, we have

$$\int_{t_0}^{t_1} |\omega(u(t), y(t))| dt < \infty. \quad (2)$$

Dissipativity and passivity are energy-based notions that characterize a dynamical system by its input/output behavior. A system is dissipative if the increase in the system's stored energy is less than the entire energy supplied to the system. The energy supplied to the system is defined by the supply function, and the energy stored in the system is defined by the storage function.

**Definition 2:** (see [38]) System  $G$  is dissipative with respect to the well-defined supply rate  $\omega(u(t), y(t))$ , if there exists a non-negative storage function  $V(x) : X \rightarrow \mathbb{R}^+$  such that, for

all  $t_0, t_1$ , where  $t_1 \geq t_0$ , and all solutions  $x(t) \in X$ ,  $u(t) \in U$ , and  $y(t) \in Y$  of the dynamical system

$$V(t_1) - V(t_0) \leq \int_{t_0}^{t_1} \omega(u(t), y(t)) dt \quad (3)$$

is satisfied. If the storage function is differentiable, we have

$$\dot{V}(t) \leq \omega(u(t), y(t)) \quad \forall t \geq 0. \quad (4)$$

**Definition 3:** (see [36]) As a special case of dissipativity, system  $G$  is called passive, if there exists a non-negative storage function  $V(x) : X \rightarrow \mathbb{R}^+$  such that

$$V(t_1) - V(t_0) \leq \int_{t_0}^{t_1} u^T(t) y(t) dt \quad (5)$$

is satisfied for all  $t_0, t_1$ , where  $t_1 \geq t_0$ , and all solutions  $x(t) \in X$ ,  $u(t) \in U$ , and  $y(t) \in Y$  of the dynamical system.

**Definition 4:** (see [37]) System  $G$  is considered to be input feed-forward output feedback passive (IF-OFP), if it is dissipative with respect to the well-defined supply rate

$$\omega(u, y) = u^T y - \rho y^T y - \nu u^T u \quad (6)$$

for some  $\rho, \nu \in \mathbb{R}$ .

The IF-OFP property presents a more general form for the concept of passivity. Based on Definition 4, we can denote an IF-OFP system with its passivity indices.  $\nu$  is called the input passivity index, and  $\rho$  is called the output passivity index. If only  $\nu > 0$ , then the system is *input strictly passive* (ISP); if only  $\rho > 0$ , then the system is *output strictly passive* (OSP). If  $\nu > 0$  and  $\rho > 0$ , then the system is *very strictly passive*.

**Lemma 1:** (see [39]) Without loss of generality, the domain of  $\rho$  and  $\nu$  in IF-OFP systems is  $\Pi = \Pi_1 \cup \Pi_2$  with  $\Pi_1 = \{\rho, \nu \in \mathbb{R} | \rho\nu < \frac{1}{4}\}$  and  $\Pi_2 = \{\rho, \nu \in \mathbb{R} | \rho\nu = \frac{1}{4} \text{ with } \rho \geq 0\}$ .

Passivity indices are a means to measure the shortage or excess of passivity in dynamical systems [37] and are useful in passivity-based analysis and control of systems [40], [41]. Passivity indices are useful for analyzing the performance of systems as well [41]. Furthermore, passivity is compositional. This means that the passive feedback or feed-forward interconnection of two passive subsystems is also passive. Additionally, the feedback interconnection of two OSP systems is OSP. And the feed-forward interconnection of two ISP systems is ISP [37]. A positive value for either one of the passivity indices indicates an excess in passivity; a negative value for either one of the passivity indices indicates a shortage in passivity. An excess of passivity in one system can compensate for the shortage of passivity in another system, leading to a passive and stable feedback or feed-forward interconnection [42].

**Theorem 1:** (see [37]) Consider a feedback interconnection of two IF-OFP systems, and suppose that the storage function for each subsystem satisfies the following inequality:

$$\dot{V}_i \leq u_i^T y_i - \rho_i y_i^T y_i - \nu_i u_i^T u_i, \text{ for } i = 1, 2. \quad (7)$$

Then, the closed-loop interconnection is finite-gain  $L_2$ -stable if

$$\rho_1 + \nu_2 > 0, \rho_2 + \nu_1 > 0. \quad (8)$$

**Definition 5:** (see [37]) System  $G$  is called finite-gain  $L_2$ -stable, if for the smallest possible positive gain  $\gamma$ , and  $\forall u(t) \in U$ , a  $\beta$  exists such that over the time interval  $[0, \tau]$  and for any positive  $\tau$ , we have

$$\|y_\tau\|_{L_2} \leq \gamma \|u_\tau\|_{L_2} + \beta. \quad (9)$$



Here,  $\|y_\tau\|_{L_2}$  and  $\|u_\tau\|_{L_2}$  represent the  $L_2$ -norm of truncated signals over the time interval  $[0, \tau]$ . For instance

$$\|y_\tau\|_{L_2} = \sqrt{\int_0^\tau y^T(t)y(t)dt}.$$

### III. PROBLEM STATEMENT

As shown in Fig. 1, we are exploring the interconnection of two IF-OFP systems. The main plant has the passivity levels  $\rho_p$  and  $\nu_p$ , and the controller has the passivity levels  $\rho_c$  and  $\nu_c$  (there are many computational methods such as LMI methods for linear systems and sum of squares methods for nonlinear systems to determine passivity indices of dynamical systems [37]). We consider a large class of systems where the indices can take positive or negative values indicating the extent that each subsystem is passive or nonpassive. Our set-up will guarantee finite-gain  $L_2$ -stability and performance improvement for the NCS framework shown in Fig. 1 in the presence of signal quantization, time delays, packet dropouts, and event-triggering conditions. This is reached based on the main ideas behind dissipativity and passivity given in Definitions 2–4.

The set-up is event-triggered on both sides of the network. The inputs to the controller and plant are held constant based on the last value received from the network. This structure is commonly used in the literature to analyze the behavior of networked interconnections, as it can capture different NCS configurations [1]. Depending on application,  $w_1(t)$  may be a bounded reference input or band-limited external disturbance on the plant's side. Our approach is based on bounded-input bounded-output stability, and as long as  $w_1(t)$  is bounded, the entire event-triggered networked system will be  $L_2$ -stable.  $w_2(t) \neq 0$  as an external disturbance on the controller's side has only been considered for examining the robustness of the triggering condition against external disturbance in Section V. We prove stability by considering  $w_2(t) = 0$ , as shown in Fig. 1. We consider simple triggering conditions for both sides

$$\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2 \quad 0 < \delta_p \leq 1 \quad (10)$$

$$\|e_c(t)\|_2^2 > \delta_c \|y_c(t)\|_2^2 \quad 0 < \delta_c \leq 1. \quad (11)$$

The triggering mechanisms shown in Fig. 1 are representing the situations in which new information is sent every time a violation of the triggering condition occurs. An event-detector is located on the output of the plant to monitor the behavior of plant's output. An updated measure of  $y_p$  is sent to the communication network when the error between the last information sent ( $y_p(t_{p_k})$ ) and the current one  $e_p(t) = y_p(t) - y_p(t_{p_k})$  (for  $t \in [t_{p_k}, t_{p_{k+1}})$ ) exceeds a predetermined threshold established by the designer based on the relation given in (10) and the design parameter  $\delta_p$ . A similar set-up is also presented on the controller's output. An updated measure of  $y_c$  is sent to the communication network when the error between the last information sent from the controller ( $y_c(t_{c_k})$ ) and the current one  $e_c(t) = y_c(t) - y_c(t_{c_k})$  (for  $t \in [t_{c_k}, t_{c_{k+1}})$ ) exceeds the threshold value based on the relation given in (11) and the design parameter  $\delta_c$ . At instances for which the triggering condition is met, and new information is successfully exchanged, the error is set back to zero:  $e_p(t_{p_{k+1}}) = 0$  or  $e_c(t_{c_{k+1}}) = 0$ . These simple triggering conditions will facilitate the design process by making it easier for the designer to understand and analyze the tradeoffs among performance,  $L_2$ -stability, channel utilization,

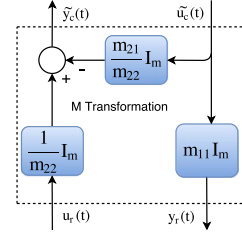


Fig. 2. Proposed  $M$ -transformation matrix.

and passivity levels of subsystems. Additionally, for the design of the conditions mentioned above, we do not need to know the exact dynamical models for each subsystem, and only access to each subsystem's output is sufficient. In [43], we showed the relationship among these event-triggering conditions and  $QSR$ -dissipativity and stability, and in this paper, we will utilize some of the former results as building blocks for our new findings. We show the process for determining  $\delta_p$  based on the transformation matrix, and  $\delta_c$  is chosen based on our previous work on this topic given in [43, Th. 3]. From (10), (11), and by taking the square root of the terms and moving terms to one side, one may have

$$\|y_p(t_{p_k})\|_2 \leq (1 + \sqrt{\delta_p}) \|y_p(t)\|_2 \quad \forall t \in [t_{p_k}, t_{p_{k+1}}) \quad (12)$$

$$\|y_c(t_{c_k})\|_2 \leq (1 + \sqrt{\delta_c}) \|y_c(t)\|_2 \quad \forall t \in [t_{c_k}, t_{c_{k+1}}). \quad (13)$$

The hold blocks function as zero-order holders.  $Q_p$  and  $Q_c$  represent the signal quantization blocks in our framework. Quantization occurs right before information is sent over to the communication network. We have considered a large class of quantizers called passive quantizers. This class includes many different methods of quantization such as logarithmic quantizers and midtread uniform quantizers. More details on this type of quantizers may be found in [12], [44], and [45]. A quantizer is passive, if its input  $u$  and output  $y$  satisfy the following relation:

$$au^T(t)u(t) \leq u^T(t)y(t) \leq bu^T(t)u(t)$$

where  $y(t) = Q(u(t))$  represents the output of the quantizer and  $0 \leq a \leq b < \infty$ . As shown in Fig. 1, we have

$$Q_p : a_p y_r^T(t) y_r(t) \leq y_r^T(t) y_{q_p}(t) \leq b_p y_r^T(t) y_r(t)$$

$$Q_c : a_c y_c^T(t_{c_k}) y_c(t_{c_k}) \leq y_c^T(t_{c_k}) y_{q_c}(t_{c_k}) \leq b_c y_c^T(t_{c_k}) y_c(t_{c_k})$$

where  $b_c > a_c \geq 0$  and  $b_p > a_p \geq 0$ , and  $y_{q_c}(t) = Q_c(y_c(t_{c_k}))$  and  $y_{q_p}(t) = Q_p(y_r(t))$  denote the relation between input and output of each quantizer. The quantizer on the controller's side directly depends on  $y_c(t_{c_k})$ , and due to the presence of the transformation matrix on the plant's side, the quantizer on the plant's side depends on  $y_p(t_{p_k})$  through  $y_r^T(t)$  (see Figs. 1 and 2). From the relationship between the input and output of a passive quantizer, we can conclude that

$$\|y_{q_p}(t)\|_2^2 \leq b_p^2 \|y_r(t)\|_2^2 \quad (14)$$

$$\|y_{q_c}(t)\|_2^2 \leq b_c^2 \|y_c(t_{c_k})\|_2^2. \quad (15)$$

It is known that time delays in NCSs degrade the performance of interconnections. Most approaches in the literature can be divided into two categories: delay-dependent and delay-independent methods [46], [47]. Here, we propose a

delay-independent input–output stability approach, where we do not assume that the length of time delays is known.  $T_1(t)$  represents the time-varying or constant network-induced delay from the plant to controller.  $T_2(t)$  represents the time-varying or constant network-induced delay from the controller to plant. We assume that there is no bound on admissible network-induced delays, namely, they may be larger than interevent time intervals. However, we have considered a causality assumption (a bounded rate of change) for time delays meaning

$$0 \leq \left| \frac{dT_1(t)}{dt} \right| \leq d_1 < 1, \quad 0 \leq \left| \frac{dT_2(t)}{dt} \right| \leq d_2 < 1. \quad (16)$$

These conditions simply mean that the time delays cannot grow faster than time itself. The relations in (16) stand for both increasing or decreasing time delays.

The idea of implementing a matrix transformation was adopted from the research done in the field of communication theory where similar methods are used to overcome the negative effects of time delays in interconnections [29], [48], and in control theory to preserve passivity [49] or to passivate nonpassive systems [40], [41], [50]. However, the authors in these works do not consider quantization in conjunction with time delays. They assume constant time delays, or in the case where they are dealing with NCS, they consider continuous or periodic information exchange between subsystems. The  $M$ -transformation matrix on the plant's side of communication links can be looked at as a simple local controller consisting of only simple gains

$$\begin{bmatrix} y_r(t) \\ u_r(t) \end{bmatrix} = M \begin{bmatrix} \tilde{u}_c(t) \\ \tilde{y}_c(t) \end{bmatrix} = \begin{bmatrix} m_{11}I_m & 0I_m \\ m_{21}I_m & m_{22}I_m \end{bmatrix} \begin{bmatrix} \tilde{u}_c(t) \\ \tilde{y}_c(t) \end{bmatrix} \quad (17)$$

where  $I_m \in R^{m \times m}$ . As shown in Fig. 1, by utilizing the  $M$ -transformation matrix, we will seek to transform the controller  $G_c$  and communication network into a new IF-OFP system  $\tilde{G}_c$  with input and output passivity indices  $\tilde{\rho}_c$  and  $\tilde{\nu}_c$ .  $\tilde{u}_c(t)$  and  $\tilde{y}_c(t)$  represent the input and output of  $\tilde{G}_c$ .  $y_r(t)$  and  $u_r(t)$  represent the information sent to and received from the communication links on the plant's side. As shown in Fig. 1,  $\tilde{G}_c$  receives its input from the event detector on the plant's side. The transformation matrix gives the networked control interconnection a distributed control architecture where the complicated control actions and computations are still being processed at the remote controllers. Our intention is to design the  $M$ -transformation matrix such that the negative effects of quantization and time delays on both sides are overcome, and the entire system is stable and robust.

Our proposed design will show a range of allowed values for the entries in the transformation matrix that guarantees stability. We show a relationship among these values, robustness, and the number of allowed consecutive packet dropouts. Hence, the designer will be able to make decisions based on application, robustness and performance criteria, reliability of communication links, and efficient utilization of the shared band-limited communication network. We show that  $L_2$ -stability, robustness, and communication rate depend on the subsystem's passivity levels, design entries in  $M$ -transformation matrix, and flexibility of our triggering conditions. It will be shown that more passive systems with larger passivity indices have better performance, are more robust, and utilize the network more efficiently.

## IV. MAIN RESULTS

This section will first outline the requirements for achieving finite-gain  $L_2$  stability in our networked control framework. Next, we design the proposed  $M$ -transformation matrix that will deal with negative effects of time delays and signal quantization, and meet the stability requirements. These results will include a range of allowed choices for the transformation matrix and triggering conditions that can both stabilize and enhance performance. Accordingly, one can select the design parameters given the performance and application criteria, reliability of communication links (see Section V), and desired communication rate.

### A. Finite-Gain $L_2$ -Stability Analysis of the Proposed Networked Control Design

**Theorem 2:** Consider the networked interconnection of two input feed-forward output feedback passive systems  $G_p$  and  $\tilde{G}_c$  in Fig. 1 with respective passivity indices of  $\nu_p$ ,  $\rho_p$ ,  $\tilde{\nu}_c$ , and  $\tilde{\rho}_c$ . Let the event instance  $t_{p_k}$  be explicitly determined by the triggering condition  $\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2$  where  $\delta_p \in (0, 1]$ , and  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$  and  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$ , where  $\alpha, \gamma > 0$  and

$$\beta(\tilde{\nu}_c) = \begin{cases} \rho_p - \frac{\delta_p \alpha}{2} & \text{if } \tilde{\nu}_c \geq 0 \\ \rho_p + 2\tilde{\nu}_c - \delta_p \left( \frac{\alpha}{2} - 2\tilde{\nu}_c \right) & \text{if } \tilde{\nu}_c < 0 \end{cases}$$

then the event-triggered networked control system is finite-gain  $L_2$ -stable from the input  $w_1(t)$  to output  $y_p(t)$ .

**Proof:** Given the systems  $G_p$  with passivity indices  $\nu_p$ ,  $\rho_p$ , and  $\tilde{G}_c$ , which is representing the transformed system encompassing the controller, network connections, and the  $M$ -transformation matrix with passivity indices  $\tilde{\nu}_c$  and  $\tilde{\rho}_c$ , there exist  $V_p(t)$  and  $\tilde{V}_c(t)$  such that

$$\begin{aligned} \dot{V}_p(t) &\leq u_p^T(t)y_p(t) - \nu_p u_p^T(t)u_p(t) - \rho_p y_p^T(t)y_p(t) \\ \dot{\tilde{V}}_c(t) &\leq \tilde{u}_c^T(t)\tilde{y}_c(t) - \tilde{\nu}_c \tilde{u}_c^T(t)\tilde{u}_c(t) - \tilde{\rho}_c \tilde{y}_c^T(t)\tilde{y}_c(t). \end{aligned}$$

According to the set-up shown in Fig. 1, the following relationships stand for any  $t \in [t_{p_k}, t_{p_{k+1}})$ :

$$\begin{aligned} u_p(t) &= w_1(t) - \tilde{y}_c(t), \quad e_p(t) = y_p(t) - y_p(t_{p_k}) \\ \tilde{u}_c(t) &= y_p(t_{p_k}) = y_p(t) - e_p(t) \end{aligned}$$

with the triggering condition:  $\langle e_p, e_p \rangle > \delta_p \langle y_p, y_p \rangle$ , where  $0 < \delta_p \leq 1$ . We consider the storage function  $V(t) = V_p(t) + \tilde{V}_c(t)$  for the entire interconnection

$$\begin{aligned} \dot{V}(t) &= \dot{V}_p(t) + \dot{\tilde{V}}_c(t) \\ &\leq u_p^T(t)y_p(t) - \nu_p u_p^T(t)u_p(t) - \rho_p y_p^T(t)y_p(t) \\ &\quad + \tilde{u}_c^T(t)\tilde{y}_c(t) - \tilde{\nu}_c \tilde{u}_c^T(t)\tilde{u}_c(t) - \tilde{\rho}_c \tilde{y}_c^T(t)\tilde{y}_c(t). \end{aligned}$$

For any  $t \in [t_{p_k}, t_{p_{k+1}})$ , it holds that  $u_p(t) = w_1(t) - \tilde{y}_c(t)$ , and  $\tilde{u}_c(t) = y_p(t_{p_k}) = y_p(t) - e_p(t)$ . Consequently, one has

$$\begin{aligned} \dot{V}(t) &\leq (w_1(t) - \tilde{y}_c(t))^T y_p(t) + (y_p(t) - e_p(t))^T \tilde{y}_c(t) \\ &\quad - \nu_p (w_1(t) - \tilde{y}_c(t))^T (w_1(t) - \tilde{y}_c(t)) - \rho_p y_p^T(t) y_p(t) \\ &\quad - \tilde{\nu}_c (y_p(t) - e_p(t))^T (y_p(t) - e_p(t)) - \tilde{\rho}_c \tilde{y}_c^T(t) \tilde{y}_c(t) \\ &= w_1^T(t) y_p(t) - \nu_p w_1^T(t) w_1(t) - (\rho_p + \tilde{\nu}_c) y_p^T(t) y_p(t) \\ &\quad - (\tilde{\rho}_c + \nu_p) \tilde{y}_c^T(t) \tilde{y}_c(t) - e_p^T(t) \tilde{y}_c(t) - \tilde{\nu}_c e_p^T(t) e_p(t) \\ &\quad + 2\nu_p w_1^T(t) \tilde{y}_c(t) + 2\tilde{\nu}_c y_p^T(t) e_p(t). \end{aligned}$$

It can be shown that  $-e_p^T(t) \tilde{y}_c(t) = -(\frac{\sqrt{\alpha} e_p(t)}{\sqrt{2}} + \frac{\tilde{y}_c(t)}{\sqrt{2\alpha}})^2 + \frac{\alpha}{2} e_p^T(t) e_p(t) + \frac{1}{2\alpha} \tilde{y}_c^T(t) \tilde{y}_c(t)$  where  $\alpha > 0$ , and  $w_1^T(t) y_p(t) = -(\sqrt{\gamma} w_1(t) - \frac{y_p(t)}{2\sqrt{\gamma}})^2 + \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma}$  where  $\gamma > 0$ . Additionally, it stands that  $e_p^T(t) e_p(t) \leq \delta_p y_p^T(t) y_p(t)$  for all  $t \in [t_{p_k}, t_{p_{k+1}})$  and  $2\nu_p w_1^T(t) \tilde{y}_c(t) \leq |\nu_p| w_1^T(t) w_1(t) + |\nu_p| \tilde{y}_c^T(t) \tilde{y}_c(t)$ . Simplifying further, one may have

$$\begin{aligned} \dot{V}(t) &\leq \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma} \\ &\quad - \nu_p w_1^T(t) w_1(t) + |\nu_p| w_1^T(t) w_1(t) + |\nu_p| \tilde{y}_c^T(t) \tilde{y}_c(t) \\ &\quad - \left( \rho_p + \tilde{\nu}_c - \frac{\delta_p \alpha}{2} \right) y_p^T(t) y_p(t) + 2\tilde{\nu}_c y_p^T(t) e_p(t) \\ &\quad - \left( \tilde{\rho}_c + \nu_p - \frac{1}{2\alpha} \right) \tilde{y}_c^T(t) \tilde{y}_c(t) - \tilde{\nu}_c e_p^T(t) e_p(t). \end{aligned}$$

Given (10) and if  $\tilde{\nu}_c \geq 0$ , one can substitute the relationship  $2\tilde{\nu}_c y_p^T(t) e_p(t) \leq |\tilde{\nu}_c| y_p^T(t) y_p(t) + |\tilde{\nu}_c| e_p^T(t) e_p(t)$  back into the above-mentioned equation to have

$$\begin{aligned} \dot{V}(t) &\leq \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma} - \nu_p w_1^T(t) w_1(t) \\ &\quad + |\nu_p| w_1^T(t) w_1(t) - \left( \rho_p - \frac{\delta_p \alpha}{2} \right) y_p^T(t) y_p(t) \\ &\quad - \left( \tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} \right) \tilde{y}_c^T(t) \tilde{y}_c(t) \end{aligned}$$

if  $\tilde{\nu}_c < 0$ , given that  $2\tilde{\nu}_c y_p^T(t) e_p(t) \leq |\tilde{\nu}_c| y_p^T(t) y_p(t) + |\tilde{\nu}_c| e_p^T(t) e_p(t)$ ,  $|\tilde{\nu}_c| e_p^T(t) e_p(t) \leq \delta_p |\tilde{\nu}_c| y_p^T(t) y_p(t)$ , and  $-\tilde{\nu}_c e_p^T(t) e_p(t) \leq \delta_p |\tilde{\nu}_c| y_p^T(t) y_p(t)$  for any  $t \in [t_{p_k}, t_{p_{k+1}})$ , one has

$$\begin{aligned} \dot{V}(t) &\leq \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma} \\ &\quad - \nu_p w_1^T(t) w_1(t) + |\nu_p| w_1^T(t) w_1(t) \\ &\quad - \left( \rho_p + 2\tilde{\nu}_c - \delta_p \left( \frac{\alpha}{2} - 2\tilde{\nu}_c \right) \right) y_p^T(t) y_p(t) \\ &\quad - \left( \tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} \right) \tilde{y}_c^T(t) \tilde{y}_c(t). \end{aligned}$$

Given  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$ , one may show

$$\begin{aligned} \dot{V}(t) &\leq (\gamma + |\nu_p| - \nu_p) w_1^T(t) w_1(t) \\ &\quad - \left( \beta(\tilde{\nu}_c) - \frac{1}{4\gamma} \right) y_p^T(t) y_p(t) \end{aligned}$$

where

$$\beta(\tilde{\nu}_c) = \begin{cases} \rho_p - \frac{\delta_p \alpha}{2} & \text{if } \tilde{\nu}_c \geq 0 \\ \rho_p + 2\tilde{\nu}_c - \delta_p \left( \frac{\alpha}{2} - 2\tilde{\nu}_c \right) & \text{if } \tilde{\nu}_c < 0. \end{cases}$$

If  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$ , we can integrate from both sides over the time interval  $[0, \tau]$  to have

$$\begin{aligned} \int_0^\tau \dot{V}(t) dt + \left( \beta(\tilde{\nu}_c) - \frac{1}{4\gamma} \right) \int_0^\tau y_p^T(t) y_p(t) dt \\ \leq (\gamma + |\nu_p| - \nu_p) \int_0^\tau w_1^T(t) w_1(t) dt. \end{aligned}$$

Given that  $V(x) \geq 0$ , we may take the square root of the both sides and simplify further to have

$$\|y_{p\tau}\|_{L_2} \leq \left( \frac{\gamma + |\nu_p| - \nu_p}{\beta(\tilde{\nu}_c) - \frac{1}{4\gamma}} \right) \|w_{1\tau}\|_{L_2} + \sqrt{\frac{V(0)}{\beta(\tilde{\nu}_c) - \frac{1}{4\gamma}}}$$

which leads to the definition of finite-gain  $L_2$ -stability and proves the theorem ( $y_{p\tau}$  and  $w_{1\tau}$  represent the integral terms of signals  $y_p(t)$  and  $w_1(t)$  over the time interval  $[0, \tau]$ ). ■

This proof reveals an interesting relationship between passivity indices and finite-gain  $L_2$ -stability. Namely, larger values of  $\beta(\tilde{\nu}_c)$  lead to tighter upper bounds on the output. Relating this to passivity indices in  $\beta(\tilde{\nu}_c)$ , it is revealed that a larger output passivity index  $\rho_p$  leads to a smaller  $L_2$  gain or a tighter upper bound for the output. Similarly, there is a direct relation between the plant's input passivity index and the interconnection's  $L_2$  gain. Namely, a positive input passivity index  $\nu_p$  leads to a tighter upper bound on the output compared to a negative input passivity index. One can look at this in terms of performance as well. For instance, if the purpose of the event-triggered NCS is to follow a trajectory, then a more passive system with larger passivity indices will have a smaller gain and a better tracking performance.

## B. Proposed Design for the $M$ -Transformation Matrix

**Theorem 3:** Consider the event-triggered networked control system shown in Fig. 1. Let the quantizers be passive quantizers,  $\rho_c > 0$ , and network-induced time-varying delays from the plant to controller ( $T_1(t)$ ) and the controller to plant ( $T_2(t)$ ) meet the causality conditions

$$0 \leq \left| \frac{dT_1(t)}{dt} \right| \leq d_1 < 1 \quad \text{and} \quad 0 \leq \left| \frac{dT_2(t)}{dt} \right| \leq d_2 < 1.$$

Let the triggering instances on the plant's and controller's sides be explicitly determined by the following relations:

$$\begin{aligned} \|e_p(t)\|_2^2 &> \delta_p \|y_p(t)\|_2^2 \quad \text{where } \delta_p \in (0, 1] \\ \|e_c(t)\|_2^2 &> \delta_c \|y_c(t)\|_2^2 \quad \text{where } \delta_c \in (0, 1] \end{aligned}$$

if the entries of the  $M$ -transformation matrix are selected such that

$$m_{22}^2 > \frac{(\frac{1}{2\alpha} + |\nu_p| - \nu_p)(2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2))}{\rho_c},$$

$$\text{where } \tilde{\rho}_c = \frac{\rho_c m_{22}^2}{2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)}.$$

$$|m_{21}| = \frac{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)}{\rho_c |m_{22}|}, \text{ where } m_{21}m_{22} < 0.$$

$$\tilde{\nu}_c = \frac{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)}{2\rho_c m_{22}^2} - \left(\frac{1}{2\rho_c} + |\nu_c|\right) b_p^2(1 + d_1)m_{11}^2,$$

$$\text{where } \tilde{\rho}_c \tilde{\nu}_c < \frac{1}{4}$$

to meet the conditions given in Theorem 2, namely,  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$  and  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$ , where  $\alpha, \gamma > 0$  and

$$\beta(\tilde{\nu}_c) = \begin{cases} \rho_p - \frac{\delta_p \alpha}{2} & \text{if } \tilde{\nu}_c \geq 0 \\ \rho_p + 2\tilde{\nu}_c - \delta_p(\frac{\alpha}{2} - 2\tilde{\nu}_c) & \text{if } \tilde{\nu}_c < 0 \end{cases}$$

then the networked control system is finite-gain  $L_2$ -stable from the input  $w_1(t)$  to output  $y_p(t)$ .

*Proof:* Given  $\rho_c > 0$ , for the IF-OFP controller, one has

$$\begin{aligned} \dot{V}_c(t) &\leq u_c^T(t)y_c(t) - \nu_c u_c^T(t)u_c(t) - \rho_c y_c^T(t)y_c(t) \\ &\leq |\nu_c|u_c^T(t)u_c(t) - \rho_c y_c^T(t)y_c(t) + u_c^T(t)y_c(t) \\ &\leq -\frac{1}{2\rho_c}(u_c(t) - \rho_c y_c(t))^2 + \frac{1}{2\rho_c}u_c^T(t)u_c(t) \\ &\quad - \frac{\rho_c}{2}y_c^T(t)y_c(t) + |\nu_c|u_c^T(t)u_c(t) \\ &\leq \left(\frac{1}{2\rho_c} + |\nu_c|\right) \|u_c(t)\|_2^2 - \frac{\rho_c}{2} \|y_c(t)\|_2^2. \end{aligned}$$

Integrating from both sides over the time interval  $t_0$  to  $t$  ( $\forall t \geq t_0 \geq 0$ ), one has

$$\begin{aligned} \Delta V_c &= V_c(t) - V_c(t_0) \\ &\leq \left(\frac{1}{2\rho_c} + |\nu_c|\right) \int_{t_0}^t \|u_c(\tau)\|_2^2 d\tau - \frac{\rho_c}{2} \int_{t_0}^t \|y_c(\tau)\|_2^2 d\tau. \end{aligned}$$

Additionally, given (14) and (16), after the zero-order hold (ZOH) block, it holds that

$$\begin{aligned} &\int_{t_0}^t \|u_c(\tau)\|_2^2 d\tau \\ &= \sum_{k=0}^N [t_{p_{k+1}} - t_{p_k} + T_1(t_{p_{k+1}}) - T_1(t_{p_k})] y_{q_p}(t_{p_k}) \\ &\leq \sum_{k=0}^N [(1 + d_1)(t_{p_{k+1}} - t_{p_k})] y_{q_p}(t_{p_k}) \\ &\leq b_p^2(1 + d_1) \int_{t_0}^t \|y_r(\tau)\|_2^2 d\tau. \end{aligned} \quad (18)$$

Similarly, given (16), after the ZOH block, one has

$$\begin{aligned} &\int_{t_0}^t \|u_r(\tau)\|_2^2 d\tau \\ &= \sum_{k=0}^N [t_{c_{k+1}} - t_{c_k} + T_2(t_{c_{k+1}}) - T_2(t_{c_k})] y_{q_c}(t_{c_k}) \\ &\leq \sum_{k=0}^N [(1 + d_2)(t_{c_{k+1}} - t_{c_k})] y_{q_c}(t_{c_k}) \\ &\leq (1 + d_2) \int_{t_0}^t \|y_{q_c}(\tau)\|_2^2 d\tau. \end{aligned} \quad (19)$$

Given (11), (13), and (15), and simplifying further leads to

$$\|y_{q_c}(t_{c_k})\|_2^2 \leq b_c^2(1 + \sqrt{\delta_c})^2 \|y_c(t)\|_2^2. \quad (20)$$

Implementing (20) in (19) leads to

$$\int_{t_0}^t \|u_r(\tau)\|_2^2 d\tau \leq b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2) \int_{t_0}^t \|y_c(\tau)\|_2^2 d\tau. \quad (21)$$

Given (18) and (21), one has

$$\begin{aligned} \Delta V_c &\leq \left(\frac{1}{2\rho_c} + |\nu_c|\right) b_p^2(1 + d_1) \int_{t_0}^t \|y_r(\tau)\|_2^2 d\tau \\ &\quad - \frac{\rho_c}{2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)} \int_{t_0}^t \|u_r(\tau)\|_2^2 d\tau. \end{aligned}$$

Additionally, based on the relation given in (17), one can show

$$y_r(t) = m_{11}\tilde{u}_c(t), \quad u_r(t) = m_{21}\tilde{u}_c(t) + m_{22}\tilde{y}_c(t).$$

This leads to

$$\Delta V_c \leq -\frac{\rho_c m_{21}m_{22}}{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)} \int_{t_0}^t \tilde{y}_c^T(\tau)\tilde{u}_c(\tau) d\tau \quad (22)$$

$$+ \left[ \left(\frac{1}{2\rho_c} + |\nu_c|\right) b_p^2(1 + d_1)m_{11}^2 \right. \quad (23)$$

$$\left. - \frac{\rho_c m_{21}^2}{2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)} \right] \int_{t_0}^t \tilde{u}_c^T(\tau)\tilde{u}_c(\tau) d\tau \quad (24)$$

$$- \frac{\rho_c m_{22}^2}{2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)} \int_{t_0}^t \tilde{y}_c^T(\tau)\tilde{y}_c(\tau) d\tau. \quad (25)$$

From Theorem 2, we know that in order for the NCS to be  $L_2$ -stable, we need  $\tilde{\rho}_c > \frac{1}{2\alpha} + |\nu_p| - \nu_p$ . From (25), we find

$$m_{22}^2 > \frac{(\frac{1}{2\alpha} + |\nu_p| - \nu_p)(2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2))}{\rho_c}.$$

Similarly, we need the following to hold for the system  $\tilde{G}_c$  to be IF-OFP (see Definition 4):

$$-\frac{\rho_c m_{21}m_{22}}{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)} = 1.$$



As a result, we find

$$|m_{21}| = \frac{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)}{\rho_c |m_{22}|}$$

where  $m_{21}m_{22} < 0$ . Now, we can calculate  $\tilde{\nu}_c$  (if we want  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$ , for a positive  $\tilde{\nu}_c$ , we will need to choose  $m_{11}$  so that  $\tilde{\nu}_c > 0$ )

$$\frac{b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2)}{2\rho_c m_{22}^2} - \left( \frac{1}{2\rho_c} + |\nu_c| \right) b_p^2(1 + d_1)m_{11}^2 = \tilde{\nu}_c$$

where  $\tilde{\rho}_c \tilde{\nu}_c < \frac{1}{4}$ , which also completes the proof. ■

*Remark 1:* As a design process,  $m_{22}$  can be selected to meet the required  $L_2$ -stability condition  $\tilde{\rho}_c > \frac{1}{2\alpha} + |\nu_p| - \nu_p$  and desired performance and robustness index. Consequently,  $m_{21}$  is easily calculated from  $m_{22}$ . And as the last step, one can calculate  $m_{11}$  for the desired  $\tilde{\nu}_c$ .

*Remark 2:*  $m_{22}$  and  $m_{11}$  can be observed to have a reciprocal relation. This means that increasing  $m_{22}$  will decrease  $m_{11}$  for a positive  $\tilde{\nu}_c$ . This is expected due to the general relationship between input passivity index and output passivity index in IF-OPF systems, i.e.,  $\tilde{\rho}_c \tilde{\nu}_c < \frac{1}{4}$ .

*Remark 3:* The results given in Theorem 3 show that if the entries in the transformation matrix  $M$  are design for a set of time delays with specific “rates of growth” ( $d_1$  and  $d_2$ ), then the same transformation matrix will also stabilize the event-triggered NCS for time delays with smaller “rates of growth” ( $d'_1 < d_1$  and  $d'_2 < d_2$ ). This means that if one has some relative knowledge on the delays existing in the communication links (such as transmission and access times delays) and some estimate on the minimum and maximum rates of change for these delays, then one can design the entire NCS for the largest possible rates of change. This will guarantee stability in the presence of possible variations in time delays.

Theorems 2 and 3 leave a lot of room for the designer to select parameters such that the desired communication rate, robustness, and performance criteria are met. The selection of  $\delta_p$  and  $\delta_c$  has a greater effect on the rate of communication needed to maintain stability. The selection of the entries of the transformation matrix, by determining the passivity indices  $\tilde{\rho}_c$  and  $\tilde{\nu}_c$ , has a greater effect on the performance and robustness of the set-up. Values of  $\rho_p$  and  $\delta_p$  together will determine the stability criteria of the NCS.  $\delta_c$  can be chosen relatively large compared to  $\delta_p$  to decrease the communication rate from the controller to plant; however, larger values of  $\delta_c$  will require larger gain blocks in the local controller on the plant's side; this can be interpreted as cost increase. This means that the selection of  $\delta_c$  has a direct effect on the communication rate and cost.

### C. Calculating the Passivity Indices for the Entire Event-Triggered NCS

As was mentioned in Section II, passivity is compositional. A passive interconnection of two passive systems is passive. Additionally, the feedback interconnection of two OSP systems is OSP and the feed-forward interconnection of two ISP systems is ISP. Moreover, passivity is closely related to stability. For example, the feedback interconnection of two passive systems is  $L_2$ -stable, if the passivity indices of subsystems meet the requirements given in Theorem 1. This motivates us to find a relationship between design parameters and the passivity indices of the entire event-triggered NCS. This is extremely useful

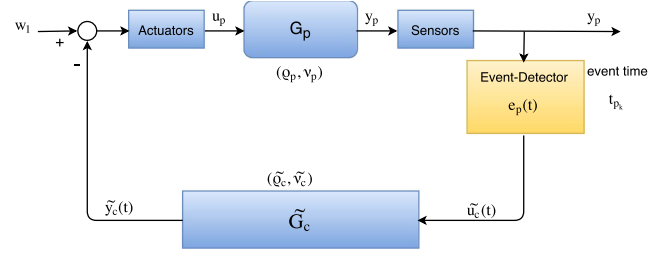


Fig. 3. Simplified model of the proposed event-triggered NCS.

in design and control of compositional cyber-physical systems. Namely, one can design larger stable passive event-triggered NCS from smaller stable passive event-triggered NCS. In this section, we will calculate the passivity indices for the entire event-triggered NCS. As a result, in addition to the design requirements given in Theorems 2 and 3 for stability, one can design the NCS to have specific passivity indices.

*Theorem 4:* Consider the networked interconnection of two IF-OPF systems  $G_p$  and  $G_c$  in Fig. 3 with respective passivity indices  $\nu_p$ ,  $\rho_p$ , and  $\tilde{\nu}_c$ ,  $\tilde{\rho}_c$ . Let the quantizers be passive quantizers and time delays follow the conditions given in (16). If the event instance  $t_{p_k}$  is explicitly determined by the triggering condition  $\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2$  where  $\delta_p \in (0, 1]$ , and the event instance  $t_{c_k}$  is explicitly determined by the triggering condition  $\|e_c(t)\|_2^2 > \delta_c \|y_c(t)\|_2^2$  where  $\delta_c \in (0, 1]$ . If the transformation matrix  $M$  is designed according to Theorem 3 so the entire event-triggered NCS is stable according to Theorem 2, then the networked interconnection is passive from  $w_1 \rightarrow y_p$  with input passivity index  $\epsilon_0$  and output passivity index  $\delta_0$ , meaning

$$\dot{V}(t) \leq w_1^T(t)y_p(t) - \epsilon_0 w_1^T(t)w_1(t) - \delta_0 y_p^T(t)y_p(t)$$

where  $\epsilon_0$  and  $\delta_0$  are selected such that

$$\epsilon_0 < \min \left( \nu_p, \frac{\nu_p(\tilde{\rho}_c - \frac{1}{2\alpha})}{\tilde{\rho}_c + \nu_p - \frac{1}{2\alpha}} \right)$$

$$0 \leq \delta_0 \leq \beta(\tilde{\nu}_c).$$

*Proof:* The proof is presented in Appendix A. ■

### V. ROBUSTNESS OF THE PROPOSED NETWORKED CONTROL DESIGN

This section will characterize the robustness of our design in two categories; robustness against external disturbance and Zeno behavior, and robustness against packet dropouts. Lower bounds for the time intervals between triggering instances (proving the lack of Zeno behavior) for the event-triggering conditions are shown. We will characterize a relationship between the performance of our proposed NCS design and undesirable external disturbances. We can define this relationship as a measure of robustness against uncertainties. We propose that our design is robust in the sense that we will not have a continuous information exchange among subunits, the event-triggered premise is met, and that our design will efficiently decrease the communication rate and stay stable.

Second, we show a tradeoff between passivity levels of subsystems in the networked control interconnection, selected triggering conditions, the entries in the proposed  $M$ -transformation matrix, and the maximum number of allowed consecutive packet losses in communication links. The problem of information loss due to packet dropouts arises when unreliable communication



channels such as wireless networks or general-purpose channels are implemented in the NCS framework. Here, we seek to characterize a relationship between design parameters and interconnection's robustness against packet dropouts. Consequently, one can select design parameters based on the reliability of the communication network and desired communication rate.

### A. Analysis on the Lower Bounds of Event-Triggering Conditions

**Proposition 1:** Consider the event-triggered networked control system shown in Fig. 1 with time delays  $T_1(t)$  and  $T_2(t)$ , and quantization blocks  $Q_p$  and  $Q_c$ . Let the plant and controller be input feed-forward output feedback passive with respective passivity indices  $\rho_p, \nu_p$ , and  $\rho_c, \nu_c$ . Consider that the transformation matrix  $M$  is implemented according to Theorem 3 to meet the stability requirements given in Theorem 2. Let the triggering instances on the plant's and controller's sides be explicitly determined by the following relations:

$$\begin{aligned} \|e_p(t)\|_2^2 &> \delta_p \|y_p(t)\|_2^2 \quad \text{where } \delta_p \in (0, 1] \\ \|e_c(t)\|_2^2 &> \delta_c \|y_c(t)\|_2^2 \quad \text{where } \delta_c \in (0, 1]. \end{aligned}$$

Assuming that the input  $w_1$  meets the following conditions:

- 1)  $\|\dot{w}_1(t)\|_2 \leq C_0$  for  $t \in [t_{p_k}, t_{p_{k+1}})$
- 2)  $\sup_{t \in [t_{p_k}, t_{p_{k+1}})} \|w_1(t)\|_2 \leq C_1$

then for any initial condition  $x_p(0)$  for the plant, the interevent time interval  $\{t_{p_{k+1}} - t_{p_k}\}$  is lower bounded by

$$t_{p_{k+1}} - t_{p_k} \geq \frac{\sqrt{\delta_p} \|y_p(t_{p_{k+1}}^-)\|_2}{\frac{C_0}{\rho_p} + \Gamma(2\theta_{k,p})(\frac{1}{\rho_p^2} + 1)(C_1 + C_2)} \quad (26)$$

where  $\sup_{t \in [t_{p_k}, t_{p_{k+1}})} \|\tilde{y}_c(t)\|_2 \leq C_2$  and

$$\Gamma(2\theta_{k,p}) = \left\| \cos^{-1} \left( \frac{\nu_p + \rho_p}{\sqrt{(1 - 4\rho_p\nu_p) + (\nu_p + \rho_p)^2}} \right) \right\|_2. \quad (27)$$

**Proof:** The proof is presented in Appendix B. ■

**Remark 4:** It can be seen that a larger triggering threshold, which depends on the choice of  $\delta_p$ , leads to larger interevent intervals. Same direct relationship holds for the output passivity index, namely, a larger  $\rho_p$  also increases the length of interevent intervals. Moreover, the relationship between passivity indices and interevent time intervals is explicitly shown in (26).

**Remark 5:** Another important aspect of the formula given in (26) is that  $\|\dot{w}_1(t)\|_2 < C_0$  and  $\sup_{t \in [t_{p_k}, t_{p_{k+1}})} \|w_1(t)\|_2 \leq C_1$  have reciprocal relationships with interevent time intervals, meaning in cases with  $w_1$  as the external disturbance to the system, the stronger the disturbance  $w_1$ , the shorter the interevent time intervals, and the triggering condition is met more frequently to compensate for the strong external disturbance. In other words, the relation given in (26) characterized the robustness of the event-triggering condition on the plant's side against external disturbance.

As it can be seen in (26), the only time  $t_{p_{k+1}}^- - t_{p_k} = 0$  is when  $\|y_p(t_{p_{k+1}}^-)\|_2 = 0$ . This can be interpreted as the case that the system is at rest or has converged to zero. In these cases, the triggering condition given in (10) is never met, and no new information is required to be sent. In other cases, such as when the

output is tracking a specific signal, we have  $\|y_p(t_{p_{k+1}}^-)\|_2 > 0$  and  $t_{p_{k+1}}^- - t_{p_k} > 0$ . However, the designed triggering condition combined with the plant output's rate of change affects the length of interevent time intervals. Once the system is getting closer to convergence, the interevent time intervals will get shorter in length: the system will need more control actions to guarantee convergence. However, in many cases, this process will be fast and short; a faster information exchange usually means more precise and efficient control actions that will result in a faster convergence. This phenomenon is common in most Zeno-behavior analyses in the literature [25], [32], [34], [51]–[53]. The idea here is that the triggering condition is never met when the right side of the equations in proofs for lack of Zeno behavior is zero, and as a result, an infinite number of triggering in a finite time interval (Zeno behavior) does not occur. We have shown this in Section VI by calculating the length of the smaller time interval between two event instances to show that Zeno behavior does not occur.

Another solution for avoiding Zeno behavior given in the literature can be seen in works such as the ones given in [35], [54], and [55], where a small positive  $\epsilon$  is introduced to the triggering condition to guarantee a positive lower bound. Same asymptotic criticism holds here as well, as usually the value of  $\epsilon$  is chosen to be very small or otherwise, this will affect the performance and stability. In Corollary 1, we show that our stability result does hold for the triggering condition  $\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2 + \epsilon$ . As a result, in practical applications, one can use this triggering condition to secure a positive lower bound, if necessary.

**Corollary 1:** Consider the networked interconnection of two input feed-forward output feedback passive systems  $G_p$  and  $\tilde{G}_c$  in Fig. 1 with respective passivity indices of  $\nu_p, \rho_p, \tilde{\nu}_c$ , and  $\tilde{\rho}_c$ . Let the event instance  $t_{p_k}$  be explicitly determined by the triggering condition  $\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2 + \epsilon$ , where  $\delta_p \in (0, 1]$ , and  $\epsilon$  is a positive number,  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$  and  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$ , where  $\alpha, \gamma > 0$  and

$$\beta(\tilde{\nu}_c) = \begin{cases} \rho_p - \frac{\delta_p \alpha}{2} & \text{if } \tilde{\nu}_c \geq 0 \\ \rho_p + 2\tilde{\nu}_c - \delta_p \left( \frac{\alpha}{2} - 2\tilde{\nu}_c \right) & \text{if } \tilde{\nu}_c < 0 \end{cases}$$

then the event-triggered networked control system is finite-gain  $L_2$ -stable from the input  $w_1(t)$  to output  $y_p(t)$ .

**Proof:** The proof follows the same line of reasoning as the proof for Theorem 2. For  $\tilde{\nu}_c \geq 0$ , one has

$$\begin{aligned} \dot{V}(t) &\leq \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma} - \nu_p w_1^T(t) w_1(t) \\ &\quad + |\nu_p| w_1^T(t) w_1(t) - \left( \rho_p - \frac{\delta_p \alpha}{2} \right) y_p^T(t) y_p(t) \\ &\quad - \left( \tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} \right) \tilde{y}_c^T(t) \tilde{y}_c(t) + \frac{\alpha \epsilon}{2} \end{aligned}$$

for  $\tilde{\nu}_c < 0$ , one has

$$\begin{aligned} \dot{V}(t) &\leq \gamma w_1^T(t) w_1(t) + \frac{y_p^T(t) y_p(t)}{4\gamma} + \left( 2\delta_p |\tilde{\nu}_c| + \frac{\alpha}{2} \right) \epsilon \\ &\quad - \nu_p w_1^T(t) w_1(t) + |\nu_p| w_1^T(t) w_1(t) \end{aligned}$$

$$\begin{aligned}
& - \left( \rho_p + 2\tilde{\nu}_c - \delta_p \left( \frac{\alpha}{2} - 2\tilde{\nu}_c \right) \right) y_p^T(t) y_p(t) \\
& - \left( \tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} \right) \tilde{y}_c^T(t) \tilde{y}_c(t).
\end{aligned}$$

In both cases, a constant positive term  $C$  has been added to the right side of the inequality ( $C = +\frac{\alpha\epsilon}{2}$  for  $\tilde{\nu}_c \geq 0$  and  $C = +(2\delta_p|\tilde{\nu}_c| + \frac{\alpha}{2})\epsilon$  for  $\tilde{\nu}_c < 0$ ). Given  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$ , and if  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$ , we can integrate from both sides over the time interval  $[0, \tau]$  to have

$$\|y_{p\tau}\|_{L_2} \leq \left( \frac{\gamma + |\nu_p| - \nu_p}{\beta(\tilde{\nu}_c) - \frac{1}{4\gamma}} \right) \|w_{1\tau}\|_{L_2} + \sqrt{\frac{V(0) + C}{\beta(\tilde{\nu}_c) - \frac{1}{4\gamma}}}$$

which proves the corollary.  $\blacksquare$

**Proposition 2:** Consider the event-triggered networked control system shown in Fig. 1 with time delays  $T_1(t)$  and  $T_2(t)$ , and quantization blocks  $Q_p$  and  $Q_c$ . Let the plant and controller be input feed-forward output feedback passive with respective passivity indices  $\rho_p, \nu_p$ , and  $\rho_c, \nu_c$ . Consider that the transformation matrix  $M$  is implemented according to Theorem 3 to meet the stability requirements given in Theorem 2. Let the triggering instances on the plant's and controller's sides be explicitly determined by the following relations:

$$\begin{aligned}
\|e_p(t)\|_2^2 &> \delta_p \|y_p(t)\|_2^2 \quad \text{where} \quad \delta_p \in (0, 1] \\
\|e_c(t)\|_2^2 &> \delta_c \|y_c(t)\|_2^2 \quad \text{where} \quad \delta_c \in (0, 1].
\end{aligned}$$

For the purpose of analyzing the proposed networked control design's robustness against external disturbance, we assume that an external disturbance  $w_2 \neq 0$ , which meets the following conditions:

- 1)  $\|\dot{w}_2(t)\|_2 \leq C'_0$  for  $t \in [t_{c_k}, t_{c_{k+1}})$
- 2)  $\sup_{t \in [t_{c_k}, t_{c_{k+1}})} \|w_2(t)\|_2 \leq C'_1$

is fed into the networked interconnection on the controller's side. Then for any initial condition  $x_p(0)$  for the plant, the interevent time interval  $\{t_{c_{k+1}} - t_{c_k}\}$  is lower bounded by

$$t_{c_{k+1}} - t_{c_k} \geq \frac{\sqrt{\delta_c} \|y_c(t_{c_{k+1}}^-)\|_2}{\frac{C'_0}{\rho_c} + \Gamma'(2\theta_{k,c}) \left( \frac{1}{\rho_c^2} + 1 \right) (C'_1 + C'_2)} \quad (28)$$

where  $\sup_{t \in [t_{c_k}, t_{c_{k+1}})} \|y_{q_p}(t)\|_2 \leq C'_2$  and

$$\Gamma'(2\theta_{k,c}) = \left\| \cos^{-1} \left( \frac{\nu_c + \rho_c}{\sqrt{(1 - 4\rho_c\nu_c) + (\nu_c + \rho_c)^2}} \right) \right\|_2. \quad (29)$$

*Proof:* The proof is presented in our internal technical report [56].  $\blacksquare$

**Remark 6:** In Propositions 1 and 2, we assumed upper bounds for the  $L_2$  norms of signals  $w_1$  and  $w_2$  and their derivatives. It is important to note that in the context of band-limited networks, these conditions are realistic and practical. Band-limited signals have finite power, an upper bound on their magnitudes, and are differentiable. Band-limited signals may be represented by their Fourier transform. As a result, the  $L_2$  norm of a band-limited signal and its derivative are bounded [57].

Contrary to other proofs, in Proposition 2, we aimed to solve the problem by assuming that there is an external disturbance

on the controller's side, namely,  $w_2 \neq 0$  in order to characterize our design's reaction to external disturbance on the controller's side. One can see that a larger triggering threshold  $\delta_c$  leads to larger interevent time intervals. Same direct relationship holds for the output passivity index, namely, a larger  $\rho_c$  also increases the length of the interevent time intervals. Moreover, the relationship between passivity indices and interevent time intervals is explicitly shown in (28). Another important aspect of the formula given in (28) is that  $\|\dot{w}_2(t)\|_2 \leq C'_0$  and  $\sup_{t \in [t_{c_k}, t_{c_{k+1}})} \|w_2(t)\|_2 \leq C'_1$  have reciprocal relationships with interevent time intervals, meaning in cases with  $w_2$  as the external disturbance to the system, the stronger the disturbance  $w_2$ , the shorter the interevent time intervals, and the triggering condition is met more frequently to compensate for the strong external disturbance. In other words, the relation given in (28) characterized the robustness of the event-triggering condition on the controller's side against external disturbances.

## B. Analysis on the Robustness of the NCS Design Against Packet Dropouts

In this section, we analyze the robustness of the interconnection presented in Fig. 1 in the case where unreliable communication links are implemented between subunits. We define robustness as the design's ability to perform as expected while tolerating a certain number of consecutive packet dropouts. Due to the continuous nature of our design, we should clarify what we mean by packet dropouts. On each side, whenever the triggering condition is met, the plant or controller attempts to send new information over the communication link, a packet is dropped if this attempt is not successful. This can be due to lossy communication links, or unsuccessful attempts at the sending or receiving side of the process, but as long as the intention to update old information is not fulfilled, we consider the packet as a lost one. This in return means that most likely, after the respective delays, the triggering condition is met again, and another packet is sent over the communication link.

The number of allowed consecutive lost packets in NCS can be seen as a measure of robustness. Namely, if the designed networked control can stay stable under a time interval of lost communication, then the specific networked control design is robust against certain number of packet dropouts or imperfect communication attempts. Finally, unlike most approaches in the literature, we do not assume a preknown maximum or minimum number of packet dropouts in the communication link, our results seek to show a design-based tradeoff between parameters and robustness of the design against loss of communication.

**Theorem 5:** Consider the event-triggered networked control system shown in Fig. 1 with time delays  $T_1(t)$  and  $T_2(t)$ , and quantization blocks  $Q_p$  and  $Q_c$ . Let the plant and controller be input feed-forward output feedback passive with respective passivity indices  $\rho_p, \nu_p$ , and  $\rho_c, \nu_c$ . Consider that the transformation matrix  $M$  is implemented according to Theorem 3 to meet the stability requirements given in Theorem 2. Let the triggering instances on the plant's and controller's sides be explicitly determined by the following relations:

$$\|e_p(t)\|_2^2 > \delta_p \|y_p(t)\|_2^2 \quad \text{where} \quad \delta_p \in (0, 1]$$

$$\|e_c(t)\|_2^2 > \delta_c \|y_c(t)\|_2^2 \quad \text{where} \quad \delta_c \in (0, 1].$$

Let the number of successive packet losses  $d_p \in \mathbb{Z}$  between two successful information exchanges  $t_{p_k}$  and  $t_{p_{k+1}}$  (for any

$t \in [t_{p_k}, t_{p_{k+1}})$  from the plant to controller satisfy

$$D_p(\tilde{\nu}_c) = \begin{cases} d_p \leq \left\lfloor \log_{(1+\sqrt{\delta_p})} \left( \sqrt{\frac{2(\rho_p - \frac{1}{4\gamma})}{\alpha}} + 1 \right) - 1 \right\rfloor & \text{if } \tilde{\nu}_c \geq 0 \\ d_p \leq \left\lfloor \log_{(1+\sqrt{\delta_p})} \left( \sqrt{\frac{2(\rho_p + 2\tilde{\nu}_c - \frac{1}{4\gamma})}{\alpha - 4\tilde{\nu}_c}} + 1 \right) - 1 \right\rfloor & \text{if } \tilde{\nu}_c < 0 \end{cases}$$

where  $\alpha, \gamma > 0$ , then the networked control system is robust against  $d_p$  consecutive packet dropouts and maintains finite-gain  $L_2$ -stability from the input  $w_1(t)$  to output  $y_p(t)$ .

*Proof:* The proof is presented in our internal technical report [56]. ■

**Remark 7:** The number of allowed packet dropouts for the communication link from the plant to controller has a direct relation with output passivity index  $\rho_p$ . This means that a networked system with a larger plant's output passivity index will stay stable for longer time intervals of lost communication.

**Remark 8:** Another point to make here is that a negative  $\tilde{\nu}_c$  can greatly harm the robustness of the NCS; however, this should not be an issue in our design, given that we can design the transformation matrix so that we have a positive  $\tilde{\nu}_c$ .

**Theorem 6:** Consider the event-triggered networked control system shown in Fig. 1 with time delays  $T_1(t)$  and  $T_2(t)$ , and quantization blocks  $Q_p$  and  $Q_c$ . Let the plant and controller be input feed-forward output feedback passive with respective passivity indices  $\rho_p, \nu_p$ , and  $\rho_c, \nu_c$ . Consider that the transformation matrix  $M$  is implemented according to Theorem 3 to meet the stability requirements given in Theorem 2. Let the triggering instances on the plant's and controller's sides be explicitly determined by the following relations:

$$\begin{aligned} \|e_p(t)\|_2^2 &> \delta_p \|y_p(t)\|_2^2 \quad \text{where } \delta_p \in (0, 1] \\ \|e_c(t)\|_2^2 &> \delta_c \|y_c(t)\|_2^2 \quad \text{where } \delta_c \in (0, 1]. \end{aligned}$$

Let the number of successive packet losses  $d_c \in \mathbb{Z}$  between two successful information exchanges  $t_{c_k}$  and  $t_{c_{k+1}}$  (for any  $t \in [t_{c_k}, t_{c_{k+1}})$ ) from the controller to plant satisfy

$$d_c \leq \left\lfloor \log_{(1+\sqrt{\delta_c})} \left( \sqrt{\frac{m_{22}^2 \rho_c}{(\frac{1}{2\alpha} + |\nu_p| - \nu_p) 2b_c^2 (1+d_2)}} + 1 \right) - 1 \right\rfloor$$

where  $\alpha, \gamma > 0$ , then the networked control system is robust against  $d_c$  consecutive packet dropouts and maintains finite-gain  $L_2$ -stability from the input  $w_1(t)$  to output  $y_p(t)$ .

*Proof:* The proof is presented in Appendix C. ■

**Remark 9:** As you can see from the results, the number of allowed consecutive packet dropouts for the communication link from the controller to plant has a direct relation with output passivity index  $\rho_c$ ; a networked control interconnection with a controller with a larger output passivity index stays stable under longer time intervals of lost communication.

**Remark 10:** Theorem 6 characterizes the relationship between  $d_c$  and  $m_{22}^2$ . This means that the designer of the  $M$ -transformation matrix, by selecting  $m_{22}^2$ , has some control over the interconnection's robustness against packet losses or unsuccessful communication attempts in the communication link from the controller to plant.

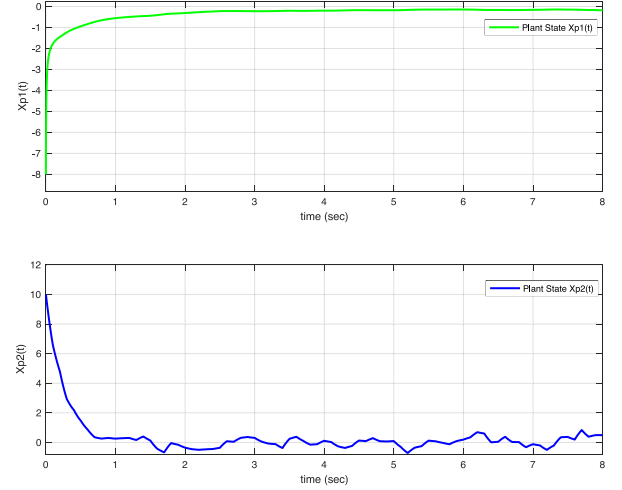


Fig. 4. States response for the proposed networked control design.

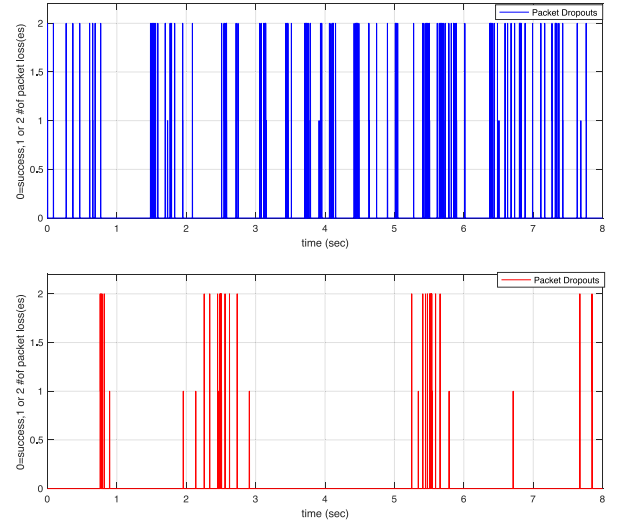


Fig. 5. Distribution of packet dropouts.

Comparing the relations given in Theorems 5 and 6, we find out that our proposed NCS design is more tolerant toward packet losses in the communication link from the controller to plant. This is similar to the observation made in Theorem 3, where the triggering condition in the communication link from the controller to plant is usually less conservative resulting in a lower communication rate, compared to the triggering condition and communication rate in the communication link from the plant to controller.

## VI. SIMULATION RESULTS

We consider an event-triggered networked interconnection of two IF-OFP systems over a band-limited networked connection that follows the structure shown in Fig. 1. We assume the following network-induced delays: an initial time delay of  $T_1(0) = 0.6$  s with an increasing rate of change of  $d_1 = 0.2$  in the communication link from the plant to controller, and an initial time delay of  $T_2(0) = 0.5$  s with an increasing rate of change of  $d_2 = 0.1$  in the communication link from the controller to plant. The quantizers used on both sides are uniform midtread quantizers with quantization level of 0.25

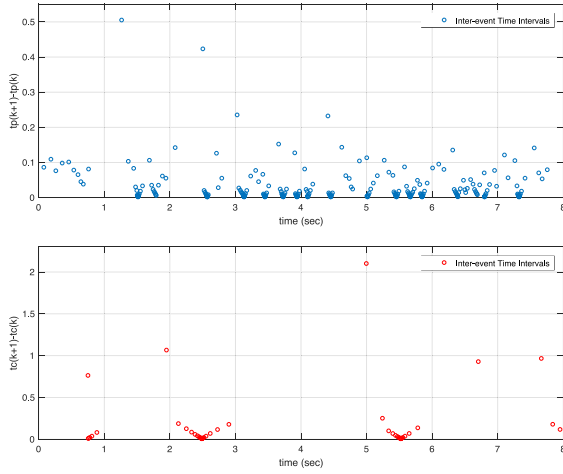


Fig. 6. Evolution of interevent time intervals.

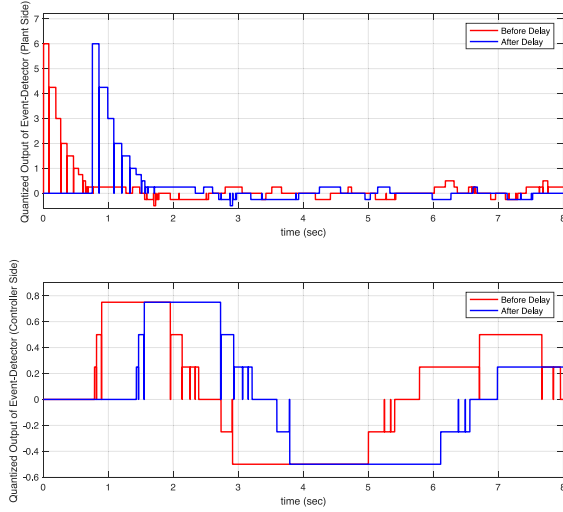


Fig. 7. Evolution of quantized plant's and controller's outputs with time delays.

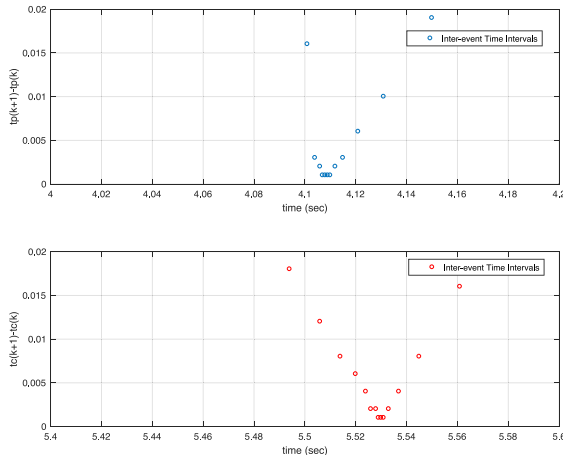


Fig. 8. Evolution of shortest interevent time intervals.

( $a_c = a_p = 0, b_c = b_p = 2$ ). Additionally, we assume imperfect network interconnections with the possibility of packet dropouts between some successful information exchanges.

The nonlinear IF-OFP plant has the dynamics

$$\dot{x}_{p1}(t) = -3x_{p1}^3(t) + x_{p1}(t)x_{p2}(t)$$

$$\dot{x}_{p2}(t) = -3.6x_{p2}(t) + 2u_p(t)$$

$$y_p(t) = x_{p2}(t)$$

with storage function  $V(x) = \frac{1}{4}x_{p2}^2$ . The plant has passivity levels  $\rho_p = 1.8$  and  $\nu_p = 0$ . The initial states of the plant are  $x_{p1}(0) = -8, x_{p2}(0) = 10$ . The nonlinear IF-OFP controller has the dynamics

$$\dot{x}_{c1}(t) = x_{c2}(t)$$

$$\dot{x}_{c2}(t) = -x_{c1}(t) - x_{c2}^3(t) + u_c(t)$$

$$y_c(t) = x_{c2}(t)$$

with storage function  $V(x) = \frac{1}{2}x_{c1}^2 + \frac{1}{2}x_{c2}^2$ . The controller has passivity levels  $\rho_c = 1$  and  $\nu_c = 0$ .

Now we will choose our design parameters according to Theorems 2 and 3. First, we select  $\delta_c = 0.25$  for the event-triggering condition on the controller's side. Next, we choose  $\gamma = \frac{1000}{4}$  and  $\alpha = 1$ , assuming that we will design the entries for the  $M$ -transformation matrix to have  $\tilde{\nu}_c > 0$ , given Theorem 3, by selecting  $\delta_p = 0.18$ , we will have  $\beta(\tilde{\nu}_c) > \frac{1}{4\gamma}$ . Our triggering conditions become

$$\|e_p(t)\|_2^2 > 0.18\|y_p(t)\|_2^2 \text{ and } \|e_c(t)\|_2^2 > 0.25\|y_c(t)\|_2^2.$$

Next, we need to select  $m_{22}^2$  such that  $\tilde{\rho}_c + \nu_p - |\nu_p| - \frac{1}{2\alpha} > 0$  where the relationship between  $\tilde{\rho}_c$  and  $m_{22}^2$  is given in Theorem 3. We choose  $m_{22}^2 = 26$  or  $m_{22}^2 = 5.09$ . This gives us  $m_{21} = -1.94$ , and finally, we select  $m_{11} = 0.26$  to have the positive passivity indices  $\tilde{\nu}_c = 0.03$  and  $\tilde{\rho}_c = 1.31$ , where as expected the following holds:  $\tilde{\rho}_c \tilde{\nu}_c < \frac{1}{4}$ .

According to Theorems 5 and 6, our specific design is robust against up to two consecutive random packet dropouts between each successful information exchange in the communication link from the plant to controller  $d_p = 2 \leq \lfloor \log_{1.42}^{(\sqrt{3.59}+1)} - 1 \rfloor$  and the controller to plant  $d_c = 2 \leq \lfloor \log_{1.5}^{(\sqrt{5.90}+1)} - 1 \rfloor$ . The distribution of successful/unsuccessful attempts over the course of our experiment is depicted in Fig. 5. We can see that both communication links are robust against 2 consecutive packet dropouts, given that stability is still preserved (see Fig. 4).

The distribution of interevent times for the event-triggering conditions on the plant's side and controller's side is shown in Fig. 6. The shortest time interval between two event instances on the plant's side occurs 4.108 s into the experiment and is calculated to be 0.009 s. The shortest time interval between two event instances on the controller's side occurs at time 5.529 s and is calculated to be 0.001 s. Fig. 8 depicts these two moments.

The external disturbance signal  $w_1(t)$  applied to the plant is a band-limited Gaussian noise with the power of 0.2. The states' response for our design is shown in Fig. 4, given  $y_p(t) = x_{p2}(t)$ , we can see that the NCS is  $L_2$ -stable. Fig. 7 shows a perspective into the effects of packet dropouts and time-varying delays on the outputs from the plant to controller and the controller to plant. It is important to mention that after-delay signals depicted in Fig. 7 are actual signals across the ZOH blocks that are fed into the controller and plant, and the before-delay signals



show what the expected evolution of the output signals would have been in the absence of the network-induced delays. As our last remark, we compare Figs. 5 and 7 to see the effects of quantization; namely, while from Fig. 5 we observe that the triggering condition is met many times, from Fig. 7 we can observe that control actions received by the plant sometimes do not change. This is due to the fact that the differences between the magnitudes of updated control actions are small enough that they lie on the same quantization interval resulting in the same final quantized values.

## VII. CONCLUSION

In this paper, we proposed a novel passivity-based design framework for finite-gain  $L_2$ -stability of event-triggered NCSs. We addressed the effects of signal quantization and time delays, which are unavoidable, when information is sent over a communication network. We analyzed and characterized the robustness issues in terms of the performance of the proposed NCS framework against packet dropouts and external disturbance as well as stability conditions for our design. To the best of our knowledge, our proposed design appears to be the most comprehensive approach existing in the networked control literature, in the sense that we consider the effects of quantization, time-varying or constant time delays, and packet dropouts altogether. Additionally, we proposed a novel conic-based analysis for finding lower bounds for interevent time intervals in our event-triggered design consisting of IF-OPF subsystems to show the robustness of our design against the undesirable “Zeno” behavior. We showed that stability is not disrupted, if a positive constant is added to the triggering conditions to guarantee a positive lower bound for the interevent time intervals. Our networked control design consists of event-triggering conditions on both sides of the interconnection; this leads to a considerable decrease in the communication load imposed on the shared band-limited network. One benefit of our passivity-based approach is its sole reliance on input–output relations of subsystems. This means that we can characterize the stability conditions for a large class of systems with unknown exact dynamics or unobservable states as long as we have an access to the outputs of each subsystem. In addition, we calculated the passivity levels for the entire proposed event-triggered NCS design. This was motivated by the fact that passivity is compositional. By determining the passivity indices of the event-triggered NCS and their relationship to design parameters, one can design larger compositional event-triggered NCS from smaller passive units. Finally, our approach is design-based thereby allowing the designer to select different design parameters based on stability, desirable communication rate, reliability network, performance, and robustness requirements.

### APPENDIX A

#### PROOF FOR THEOREM 4

*Proof:*  $G_p$  and  $\tilde{G}_c$  in Fig. 3 are IF-OPF systems with respective passivity indices  $\nu_p$ ,  $\rho_p$ , and  $\tilde{\nu}_c$ ,  $\tilde{\rho}_c$

$$\begin{aligned} \dot{V}(t) &\leq u_p^T(t)y_p(t) - \nu_p u_p^T(t)u_p(t) - \rho_p y_p^T(t)y_p(t) \\ &\quad + \tilde{u}_c^T(t)\tilde{y}_c(t) - \tilde{\nu}_c \tilde{u}_c^T(t)\tilde{u}_c(t) - \tilde{\rho}_c \tilde{y}_c^T(t)\tilde{y}_c(t) \end{aligned}$$

$$\begin{aligned} &= (w_1(t) - \tilde{y}_c(t))^T y_p(t) + (y_p(t) - e_p(t))^T \tilde{y}_c(t) \\ &\quad - \nu_p (w_1(t) - \tilde{y}_c(t))^T (w_1(t) - \tilde{y}_c(t)) - \rho_p y_p^T(t)y_p(t) \\ &\quad - \tilde{\nu}_c (y_p(t) - e_p(t))^T (y_p(t) - e_p(t)) - \tilde{\rho}_c \tilde{y}_c^T(t)\tilde{y}_c(t) \\ &= w_1^T(t)y_p(t) - \nu_p w_1^T(t)w_1(t) - e_p^T(t)\tilde{y}_c(t) \\ &\quad - (\rho_p + \tilde{\nu}_c)y_p^T(t)y_p(t) - (\tilde{\rho}_c + \nu_p)\tilde{y}_c^T(t)\tilde{y}_c(t) \\ &\quad + 2\nu_p w_1^T(t)\tilde{y}_c(t) + 2\tilde{\nu}_c y_p^T(t)e_p(t) - \tilde{\nu}_c e_p^T(t)e_p(t). \end{aligned}$$

One can show that  $-e_p^T(t)\tilde{y}_c(t) = -(\frac{\sqrt{\alpha}e_p(t)}{\sqrt{2}} + \frac{\tilde{y}_c(t)}{\sqrt{2\alpha}})^2 + \frac{\alpha}{2}e_p^T(t)e_p(t) + \frac{1}{2\alpha}\tilde{y}_c^T(t)\tilde{y}_c(t)$  where  $\alpha > 0$ . Additionally, we have  $e_p^T(t)e_p(t) \leq \delta_p y_p^T(t)y_p(t)$  for all  $t \in [t_{p_k}, t_{p_{k+1}})$ . Hence, by simplifying further, implementing the results in Theorem 2, and in order to calculate passivity indices, one can have

$$\begin{aligned} \dot{V}(t) &\leq w_1^T(t)y_p(t) - \nu_p w_1^T(t)w_1(t) - \beta(\tilde{\nu}_c)y_p^T(t)y_p(t) \\ &\quad - \left(\tilde{\rho}_c + \nu_p - \frac{1}{2\alpha}\right)\tilde{y}_c^T(t)\tilde{y}_c(t) + 2\nu_p w_1^T(t)\tilde{y}_c(t) \\ &\leq w_1^T(t)y_p(t) - \epsilon_0 w_1^T(t)w_1(t) - \delta_0 y_p^T(t)y_p(t). \end{aligned}$$

Simplifying the aforementioned and moving the terms to one side, one has

$$\begin{aligned} &+ (\epsilon_0 - \nu_p)w_1^T(t)w_1(t) + (\delta_0 - \beta(\tilde{\nu}_c))y_p^T(t)y_p(t) \\ &- \left(\tilde{\rho}_c + \nu_p - \frac{1}{2\alpha}\right)\tilde{y}_c^T(t)\tilde{y}_c(t) + 2\nu_p w_1^T(t)\tilde{y}_c(t) \\ &\leq 0. \end{aligned}$$

We need to show that

$$(\delta_0 - \beta(\tilde{\nu}_c))y_p^T(t)y_p(t) + [w_1^T(t)\tilde{y}_c^T(t)]M\begin{bmatrix} w_1(t) \\ \tilde{y}_c(t) \end{bmatrix} \leq 0$$

where  $M = \begin{bmatrix} \epsilon_0 - \nu_p & \nu_p \\ \nu_p & -(\tilde{\rho}_c + \nu_p - \frac{1}{2\alpha}) \end{bmatrix}$ . It is easy to see that for the relation to hold, the followings should be met:

$$\begin{aligned} \beta(\tilde{\nu}_c) &\geq \delta_0, \quad \nu_p > \epsilon_0 \\ \nu_p \left(\tilde{\rho}_c - \frac{1}{2\alpha}\right) &\geq \epsilon_0 \left(\tilde{\rho}_c + \nu_p - \frac{1}{2\alpha}\right) \end{aligned}$$

which also proves the theorem.  $\blacksquare$

### APPENDIX B

#### PROOF FOR PROPOSITION 1

*Proof:* Our proof will be based on the conic properties of IF-OPF systems [58]. It is beneficial to point out that the input and output of system  $G_p$  in Fig. 1 with passivity indices of  $\nu_p$ ,  $\rho_p$  occupy a conic sector in the input–output space [49], [59], [60], which is defined by its center line  $\theta_z$  and its apex angle  $2\theta_{k,p}$ . More importantly, at any instance  $t$ , the input and output lie within the conic sector:

$$\theta_p(t) \in [\theta_z - \theta_{k,p}, \theta_z + \theta_{k,p}]. \quad (30)$$

We can define the relationship between the input and output with  $\theta_p(t)$  by parameterizing them in polar coordinates:

$$u_p(t) = r_p(t) \cos(\theta_p(t)), \quad y_p(t) = r_p(t) \sin(\theta_p(t)).$$

Furthermore, we have

$$\cos(2\theta_{k,p}) = \frac{\nu_p + \rho_p}{\sqrt{(1 - 4\rho_p\nu_p) + (\nu_p + \rho_p)^2}}, \quad \theta_{k,p} \in \left[0, \frac{\pi}{2}\right).$$

Since  $e_p(t) = y_p(t) - y_p(t_{p_k})$  for all  $t \in [t_{p_k}, t_{p_{k+1}})$ , we have the following for any  $t \in [t_{p_k}, t_{p_{k+1}})$ :

$$\begin{aligned} \frac{d}{dt} \|e_p(t)\|_2 &\leq \|\dot{e}_p(t)\|_2 = \|\dot{y}_p(t)\|_2 \\ &= \|\dot{r}_p(t) \sin(\theta_p(t))\|_2 \\ &\quad + \|r_p(t) \dot{\theta}_p(t) \cos(\theta_p(t))\|_2. \end{aligned}$$

Given (30), we have  $\|\dot{\theta}_p(t)\|_2 \leq \|2\theta_{k,p}\|_2$ , where

$$\begin{aligned} \|2\theta_{k,p}\|_2 &= \left\| \cos^{-1} \left( \frac{\nu_p + \rho_p}{\sqrt{(1 - 4\rho_p\nu_p) + (\nu_p + \rho_p)^2}} \right) \right\|_2 \\ &= \Gamma(2\theta_{k,p}). \end{aligned}$$

Hence, we have

$$\frac{d}{dt} \|e_p(t)\|_2 \leq \|\dot{e}_p(t)\|_2 = \|\dot{y}_p(t)\|_2 \quad (31)$$

$$= \|\dot{r}_p(t) \sin(\theta_p(t))\|_2 + \|r_p(t) \dot{\theta}_p(t) \cos(\theta_p(t))\|_2 \quad (32)$$

$$\leq \|\dot{r}_p(t) \sin(\theta_p(t))\|_2 + \Gamma(2\theta_{k,p}) \|u_p(t)\|_2. \quad (33)$$

Furthermore, we have

$$\begin{aligned} \|\dot{r}_p(t)\|_2 &= \left\| \frac{d}{dt} \left( \frac{u_p(t)}{\cos(\theta_p(t))} \right) \right\|_2 \\ &= \left\| \frac{\dot{u}_p(t) \cos(\theta_p(t)) + u_p(t) \dot{\theta}_p(t) \sin(\theta_p(t))}{\cos^2(\theta_p(t))} \right\|_2. \end{aligned}$$

Inserting the aforementioned in (33), we have

$$\begin{aligned} \frac{d}{dt} \|e_p(t)\|_2 &\leq \\ &\|\dot{u}_p(t) \tan(\theta_p(t)) + u_p(t) \Gamma(2\theta_{k,p}) \tan^2(\theta_p(t))\|_2 \\ &\quad + \Gamma(2\theta_{k,p}) \|u_p(t)\|_2. \end{aligned}$$

We have  $\rho_p > \frac{\delta_p \alpha}{2} + \frac{1}{4\gamma}$  or  $\rho_p > \delta_p \left(\frac{\alpha}{2} - 2\tilde{\nu}_c\right) - 2\tilde{\nu}_c + \frac{1}{4\gamma}$  with  $\alpha, \gamma > 0$  for the  $L_2$ -stable networked interconnection according to Theorem 2; hence, we can conclude that  $\frac{y_p(t)}{u_p(t)} = \tan(\theta_p(t)) \leq \frac{1}{\rho_p}$  according to in [37, Lemma 6.5]. Given that  $\tilde{y}_c(t) = \frac{1}{m_{22}} y_{q_c}(t) - \frac{m_{21}}{m_{22}} y_p(t_{p_k})$  takes constant piecewise values over the time interval  $[t_{p_k}, t_{p_{k+1}})$ :  $\|\tilde{y}_c(t)\|_2 = 0$  and we can denote  $\sup_{t \in [t_{p_k}, t_{p_{k+1}})} \|\tilde{y}_c(t)\|_2 \leq C_2$ . As a result, we have  $\|u_p(t)\|_2 = \|w_1(t) - \tilde{y}_c(t)\|_2 \leq \|w_1(t)\|_2 +$

$\|\tilde{y}_c(t)\|_2 \leq C_1 + C_2$ , and  $\|\dot{u}_p(t)\|_2 = \|\dot{w}_1(t)\|_2 \leq C_0$ . Hence

$$\begin{aligned} \frac{d}{dt} \|e_p(t)\|_2 &\leq \|\dot{u}_p(t)\|_2 \|\tan(\theta_p(t))\|_2 + \Gamma(2\theta_{k,p}) \|u_p(t)\|_2 \\ &\quad + \Gamma(2\theta_{k,p}) \|u_p(t)\|_2 \|\tan^2(\theta_p(t))\|_2 \\ &\leq \frac{C_0}{\rho_p} + \Gamma(2\theta_{k,p}) \left( \frac{1}{\rho_p^2} + 1 \right) (C_1 + C_2). \end{aligned}$$

We can find a positive lower bound for interevent time intervals between each two triggered instances by integrating from both sides over the time interval  $[t_{p_k}, t_{p_{k+1}})$  and with the initial condition  $e_p(t_{p_k}) = 0$  given that after each triggering instance the error is reset to zero. We will have

$$t_{p_{k+1}}^- - t_{p_k} \geq \frac{\|e_p(t_{p_{k+1}}^-)\|_2}{\frac{C_0}{\rho_p} + \Gamma(2\theta_{k,p}) \left( \frac{1}{\rho_p^2} + 1 \right) (C_1 + C_2)}.$$

The next triggering condition is met at  $t_{p_{k+1}}$  meaning  $\|e_p(t_{p_{k+1}})\|_2^2 > \delta_p \|y_p(t_{p_{k+1}})\|_2^2$ , this gives us  $\|e_p(t_{p_{k+1}}^-)\|_2 = \sqrt{\delta_p} \|y_p(t_{p_{k+1}}^-)\|_2$ :

$$t_{p_{k+1}}^- - t_{p_k} \geq \frac{\sqrt{\delta_p} \|y_p(t_{p_{k+1}}^-)\|_2}{\frac{C_0}{\rho_p} + \Gamma(2\theta_{k,p}) \left( \frac{1}{\rho_p^2} + 1 \right) (C_1 + C_2)}$$

where

$$\Gamma(2\theta_{k,p}) = \left\| \cos^{-1} \left( \frac{\nu_p + \rho_p}{\sqrt{(1 - 4\rho_p\nu_p) + (\nu_p + \rho_p)^2}} \right) \right\|_2$$

which proves the proposition.  $\blacksquare$

## APPENDIX C PROOF FOR THEOREM 6

*Proof:* First, we assume that the broadcast release times can take values  $\{n'_i\}_{i=0}^{\infty}$ . And the two successful release times before and after each group of packet dropouts are  $t_{c_k}$  and  $t_{c_{k+1}}$  where  $k$  is a positive integer. We assume  $\delta_c$  was chosen to meet the  $L_2$ -stability requirements given in Theorem 2 and the design requirements in Theorem 3. The following upper bound holds for  $(1 + \sqrt{\delta_c})$  based on

$$m_{22}^2 > \frac{(\frac{1}{2\alpha} + |\nu_p| - \nu_p)(2b_c^2(1 + \sqrt{\delta_c})^2(1 + d_2))}{\rho_c}$$

which gives us

$$\sqrt{\frac{m_{22}^2 \rho_c}{(\frac{1}{2\alpha} + |\nu_p| - \nu_p) 2b_c^2(1 + d_2)}} > 1 + \sqrt{\delta_c}$$

where  $\alpha, \gamma > 0$ .

We considered  $d_c$  as the number of packets that are lost in the communication network from the controller to plant for the time interval  $[t_{c_k}, t_{c_{k+1}})$ , so we have  $t_{c_k} = n'_0 < n'_1 < \dots < n'_{d_c} < n'_{d_c+1} = t_{c_{k+1}}$ . For any  $t \in [n'_{d_c}, n'_{d_c+1}]$ , one has

$$\begin{aligned} \|e_c(t)\|_2 &= \|y_c(t) - y_c(t_{c_k})\|_2 = \|y_c(t) - y_c(n'_0)\|_2 \\ &\leq \sum_{i=0}^{d_c-1} \|y_c(n'_{i+1}) - y_c(n'_i)\|_2 + \|y_c(t) - y_c(n'_{d_c})\|_2. \end{aligned}$$

Assuming that the triggering condition (11) holds, and that  $d_c$  is the number of packet dropouts in the time interval  $[t_{c_k}, t_{c_{k+1}})$ , for each term in the above-mentioned equation, one has

$$\|y_c(n'_{i+1}) - y_c(n'_i)\|_2 \leq \sqrt{\delta_c} \|y_c(n'_{i+1})\|_2, \quad i = 0, \dots, d_c.$$

Hence, the following is true for any  $t \in [n'_{d_c}, n'_{d_c+1}]$ :

$$\|e_c(t)\|_2 \leq \sum_{i=0}^{d_c} \sqrt{\delta_c} \|y_c(n'_{i+1})\|_2.$$

Additionally, it holds that

$$\begin{aligned} \|e_c(t_{c_{k+1}})\|_2 &= \|y_c(t_{c_{k+1}}) - y_c(n'_{d_c})\|_2 \\ &\leq \|y_c(n'_{d_c}) - y_c(n'_{d_c+1})\|_2 \\ &\leq \sqrt{\delta_c} \|y_c(n'_{d_c+1})\|_2. \end{aligned}$$

This gives us  $\|y_c(n'_{d_c})\|_2 \leq (1 + \sqrt{\delta_c}) \|y_c(n'_{d_c+1})\|_2$ , and accordingly

$$\|y_c(n'_1)\|_2 \leq (1 + \sqrt{\delta_c})^{d_c} \|y_c(n'_{d_c+1})\|_2.$$

For  $d_c$  packet dropouts, we have

$$\|e_c(t_{c_{k+1}})\|_2 \leq \sum_{i=0}^{d_c} \sqrt{\delta_c} (1 + \sqrt{\delta_c})^i \|y_c(t_{c_{k+1}})\|_2. \quad (34)$$

Equation (34) is a geometric series with the ratio of  $1 + \sqrt{\delta_c}$ , so we have the following for all  $t \in [t_{c_k}, t_{c_{k+1}})$ :

$$\begin{aligned} \|e_c(t_{c_{k+1}})\|_2 &\leq \frac{\sqrt{\delta_c} [1 - (1 + \sqrt{\delta_c})^{d_c+1}]}{1 - (1 + \sqrt{\delta_c})} \|y_c(t_{c_{k+1}})\|_2 \\ &\leq ((1 + \sqrt{\delta_c})^{d_c+1} - 1) \|y_c(t_{c_{k+1}})\|_2. \end{aligned}$$

Given the conditions in Theorems 2 and 3, the following should hold for the interconnection to stay stable:

$$(1 + \sqrt{\delta_c})^{d_c+1} - 1 < \sqrt{\frac{m_{22}^2 \rho_c}{(\frac{1}{2\alpha} + |\nu_p| - \nu_p) 2b_c^2 (1 + d_2)}}.$$

For the aforementioned to hold, we should have

$$d_c \leq \left\lceil \log_{(1 + \sqrt{\delta_c})} \sqrt{\frac{m_{22}^2 \rho_c}{(\frac{1}{2\alpha} + |\nu_p| - \nu_p) 2b_c^2 (1 + d_2)}} + 1 \right\rceil$$

for any  $t \in [t_{c_k}, t_{c_{k+1}})$ , which proves the theorem. ■

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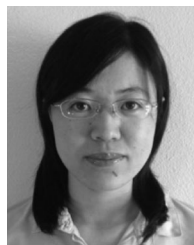


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